

❖ LangChain

1. What is a Model in LangChain

- A model in LangChain is an AI model that LangChain communicates with to generate responses, analyze text, summarize information, write code, etc.
- LangChain itself is not the AI model. It is a framework that provides tools and abstractions for working with AI models.



2. Main types of models in LangChain

- 1) LLM / Text Completion Models
- 2) Chat Models

1) LLM / Text Completion Models

- These are traditional text-generation models.

2) Chat Models

- Used for conversational AI.

Feature	LLM Model	Chat Model
Input	Usually plain text/string	Messages (System, Human, AI)
Output	Usually plain text	Usually an AIMessage
Main use	Text completion/generation	Conversations and chat applications
Example	HuggingFacePipeline	ChatOpenAI, ChatHuggingFace

3. Types of AI Models

- AI models can broadly be divided into two types:
 - 1) Closed-Source Models
 - 2) Open-Source / Open-Weight Models

1) Closed-Source Models

- Closed-source models are AI models whose **model architecture, training data, and/or model weights are not publicly available** for users to freely inspect, modify, or redistribute. They are generally accessed through an API or platform provided by the company.

Examples:

- **OpenAI** → GPT models
- **Google** → Gemini
- **Anthropic** → Claude

2) Open-Source / Open-Weight Models

- Open-source AI models are models whose important components, particularly the **model weights and often the code or related resources**, are publicly available so developers can download, run, customize, or integrate them, subject to the model's license.

Examples:

- **Hugging Face** → A platform/hub hosting many open models
- **Meta** → Llama
- **Mistral AI** → Mistral models

Closed-Source Models

Advantages	Disadvantages
1. Easy to use through APIs	1. Less control over the model
2. No need to manage servers or infrastructure	2. API usage can be costly
3. Provider handles updates, scaling, and maintenance	3. Limited access to model internals

Open-Source / Open-Weight Models

Advantages	Disadvantages
1. Can often be downloaded and run locally	1. Requires suitable hardware and computing resources
2. Can be customized or fine-tuned	2. Setup and maintenance can be more complex
3. Greater control over deployment and data	3. Performance and support can vary between models

1. LLM Model Example

- Using Google gemini Api

```
from langchain_google_genai import GoogleGenerativeAI

model = GoogleGenerativeAI(
    model="gemini-3.8-flash",
    google_api_key = "AQ.Ab8RN6LRgG4bmjJSBppac6KvJWz5xJBVoFj_7LQK1cWcU9ecPw"
)

result = model.invoke("what is python")

print(result)
```

2. Chat Model Example

- Using Google gemini Api

```
from langchain_google_genai import ChatGoogleGenerativeAI

model = ChatGoogleGenerativeAI(
    model="gemini-3.8-flash",
    google_api_key="AQ.Ab8RN6LRgG4bmjJSBppac6KvJWz5xJBVoFj_7LQK1cWcU9ecPw"
)

res = model.invoke("who is pm of india")

print(res.text)
```

3. Chat Model Example

- HuggingFace Api

```
from langchain_huggingface import ChatHuggingFace, HuggingFaceEndpoint

model = HuggingFaceEndpoint(
    repo_id="meta-llama/Llama-3.1-8B-Instruct",
    task="text-generation",
    huggingfacehub_api_token="hf_SgOuzfRPZhdJuuhdrYjbeajSmKoEvSLjj0"
)

llmmodel = ChatHuggingFace(llm=model)

result = llmmodel.invoke("Who is the prime minister of India?")

print(result.content)
```

What is Hugging Face?

- Hugging Face is an AI/ML platform and community that provides access to thousands of pre-trained models, datasets, and tools for building AI applications.
- It is especially popular for working with open-source/open-weight AI models and integrating them into applications.